

CampusTrade Frontend Test Report

Generated from real Vitest frontend test output

Run timestamp: 2026-05-05 21:22:41

Primary source command: npm run test:run

Machine-readable source: frontend/vitest-report.json

1. Run Summary

Metric	Count
Total test files	22
Total tests collected	96
Passed	94
Failed	0
Skipped	2
Pass rate	97.92%
Wall-clock duration	5.38 s

2. Per-file Breakdown

Test file	Tests	Passed	Failed	Skipped	Time (s)	Result
App.test.jsx	7	7	0	0	0.266	PASSED
sanity.test.jsx	1	1	0	0	0.019	PASSED
components/AdminRoute.test.jsx	2	2	0	0	0.061	PASSED
components/NotificationBell.test.jsx	2	2	0	0	0.284	PASSED
components/ProductCard.test.jsx	4	4	0	0	0.286	PASSED
components/ProtectedRoute.test.jsx	2	2	0	0	0.059	PASSED
components/UserCenterSidebar.test.jsx	2	2	0	0	0.170	PASSED
context/AuthContext.test.jsx	4	4	0	0	0.266	PASSED
context/UnreadContext.test.jsx	5	5	0	0	0.363	PASSED
context/WebSocketContext.test.jsx	3	3	0	0	0.035	PASSED
pages/smoke.test.jsx	17	17	0	0	0.281	PASSED
hooks/useWebSocket.test.js	7	7	0	0	0.024	PASSED
responsive/breakpoints.test.jsx	6	6	0	0	1.009	PASSED
utils/api.test.js	11	11	0	0	0.011	PASSED
utils/authRedirect.test.js	3	3	0	0	0.004	PASSED

forms/ChangePassword.test.jsx	2	2	0	0	0.727	PASSED
forms/EditProduct.test.jsx	2	2	0	0	0.304	PASSED
forms/ForgotPassword.test.jsx	2	2	0	0	0.790	PASSED
forms/Login.test.jsx	3	3	0	0	1.485	PASSED
forms/PublishProduct.test.jsx	5	3	0	2	1.749	PASSED WITH SKIPS
forms/Register.test.jsx	4	4	0	0	3.135	PASSED
forms/ResetPassword.test.jsx	2	2	0	0	0.797	PASSED

3. Coverage Areas

The frontend suite validates React client-side behaviour in a mocked browser environment. The main coverage areas extracted from the implemented test files are summarised below.

Coverage area	What was validated
Forms	Login, register, forgot password, reset password, change password, publish product, and edit product validation flows.
Components	Route guards, notification display, product card behaviour, and user-centre navigation rendering.
Contexts	Authentication restoration, unread-count state, and WebSocket provider behaviour.
Utilities	API base resolution, token and user storage rules, auth fetch headers, and login redirect behaviour.
Custom hook	WebSocket connection, reconnection, heartbeat, message receipt, and disconnect behaviour.
Smoke tests	Safe rendering of key pages such as Home, Login, Register, ProductDetail, PublishProduct, Chat, MyOrders, and admin pages.
Responsive checks	Representative mobile, tablet, and desktop breakpoint rendering for Home and ProductCard.

4. Skipped Tests

Two tests were intentionally skipped in the current frontend suite. They are listed below together with the implementation-aligned reason.

File	Skipped test	Reason
forms/PublishProduct.test.jsx	treats zero as an invalid price	Current implementation allows 0 because validation checks $\text{price} < 0$ rather than $\text{price} \leq 0$.
forms/PublishProduct.test.jsx	requires at least one uploaded image before publishing	Current implementation does not enforce an image-upload requirement at submit time.

5. All Test Cases (parsed from Vitest JSON report)

Every row below was extracted from the Vitest JSON reporter. The status column reflects the actual outcome exported by Vitest for the current frontend run.

#	File	Test name	Status	Time (ms)
1	App.test.jsx	renders the Home page for /	PASSED	126.591
2	App.test.jsx	renders the Home page for /home	PASSED	14.358
3	App.test.jsx	renders the Login page for /login	PASSED	15.728
4	App.test.jsx	renders the Register page for /register	PASSED	16.294
5	App.test.jsx	redirects unauthenticated users from a protected route to login	PASSED	30.624
6	App.test.jsx	blocks non-admin users from admin routes as implemented	PASSED	31.306
7	App.test.jsx	falls back to home for unknown routes	PASSED	30.831
8	sanity.test.jsx	renders hello vitest	PASSED	19.285
9	components/AdminRoute.test.jsx	redirects non-admin users away from the route	PASSED	46.209
10	components/AdminRoute.test.jsx	allows admin users to view the protected content	PASSED	13.949
11	components/NotificationBell.test.jsx	renders the unread count and opens the notification list	PASSED	206.387
12	components/NotificationBell.test.jsx	shows the empty state when there are no notifications	PASSED	76.917
13	components/ProductCard.test.jsx	renders title, price, and fallback image	PASSED	107.874
14	components/ProductCard.test.jsx	shows the favoured state and can toggle it for authenticated users	PASSED	69.352

15	components/ProductCard.test.jsx	redirects unauthenticated users when they try to favourite an item	PASSED	46.376
16	components/ProductCard.test.jsx	navigates to the product detail page when the card is clicked	PASSED	61.589
17	components/ProtectedRoute.test.jsx	redirects unauthenticated users to the login page	PASSED	25.464
18	components/ProtectedRoute.test.jsx	renders child content for an authenticated user	PASSED	33.377
19	components/UserCenterSidebar.test.jsx	renders the main menu sections and links	PASSED	150.526
20	components/UserCenterSidebar.test.jsx	marks the current route as active	PASSED	19.564
21	context/AuthContext.test.jsx	defaults to unauthenticated when no stored token exists	PASSED	16.918
22	context/AuthContext.test.jsx	restores the current user when a stored token exists	PASSED	76.888
23	context/AuthContext.test.jsx	syncs storage when setUser is called manually	PASSED	17.060
24	context/AuthContext.test.jsx	clears the user on auth:logout and refreshes on auth:login	PASSED	155.245
25	context/UnreadContext.test.jsx	defaults safely and avoids requests when unauthenticated	PASSED	19.744
26	context/UnreadContext.test.jsx	loads the initial unread count when authenticated	PASSED	77.376
27	context/UnreadContext.test.jsx	updates the unread count from websocket unread_update messages	PASSED	94.762
28	context/UnreadContext.test.jsx	uses websocket read notifications when sendMessage succeeds	PASSED	79.101

29	context/UnreadContext.test.jsx	falls back to REST and updates the total unread count when websocket send fails	PASSED	92.152
30	context/WebSocketContext.test.jsx	renders the provider safely and exposes the hook value to consumers	PASSED	26.028
31	context/WebSocketContext.test.jsx	returns the websocket API from useWs inside the provider	PASSED	3.225
32	context/WebSocketContext.test.jsx	throws when useWs is used outside the provider	PASSED	5.661
33	pages/smoke.test.jsx	renders Home without crashing	PASSED	49.274
34	pages/smoke.test.jsx	renders Login without crashing	PASSED	61.079
35	pages/smoke.test.jsx	renders Register without crashing	PASSED	10.659
36	pages/smoke.test.jsx	renders ProductDetail without crashing	PASSED	39.903
37	pages/smoke.test.jsx	renders PublishProduct without crashing	PASSED	20.069
38	pages/smoke.test.jsx	renders Chat without crashing	PASSED	19.141
39	pages/smoke.test.jsx	renders MyOrders without crashing	PASSED	6.021
40	pages/smoke.test.jsx	renders MeProfile without crashing	PASSED	9.002
41	pages/smoke.test.jsx	renders MyFavorites without crashing	PASSED	3.056
42	pages/smoke.test.jsx	renders Conversations without crashing	PASSED	9.126
43	pages/smoke.test.jsx	renders OrderDetail without crashing	PASSED	5.871
44	pages/smoke.test.jsx	renders EditProduct without crashing	PASSED	6.852
45	pages/smoke.test.jsx	renders MyProducts without crashing	PASSED	2.169

46	pages/smoke.test.jsx	renders SellerProfile without crashing	PASSED	25.140
47	pages/smoke.test.jsx	renders AdminUsers without crashing	PASSED	5.090
48	pages/smoke.test.jsx	renders AdminProducts without crashing	PASSED	4.753
49	pages/smoke.test.jsx	renders AdminReview without crashing	PASSED	2.881
50	hooks/useWebSocket.test.js	creates a websocket connection when a token is available	PASSED	13.015
51	hooks/useWebSocket.test.js	does not create a websocket connection when no token is available	PASSED	1.516
52	hooks/useWebSocket.test.js	updates lastMessage when an incoming websocket message arrives	PASSED	1.984
53	hooks/useWebSocket.test.js	sendMessage sends serialized data through the open socket	PASSED	1.653
54	hooks/useWebSocket.test.js	disconnect closes the socket and prevents further sends	PASSED	1.882
55	hooks/useWebSocket.test.js	sends heartbeat pings and reconnects after an unexpected close	PASSED	1.709
56	hooks/useWebSocket.test.js	reacts to auth login and logout events as implemented	PASSED	1.786
57	responsive/breakpoints.test.jsx	renders ProductCard safely at the 'mobile' breakpoint	PASSED	111.998
58	responsive/breakpoints.test.jsx	renders ProductCard safely at the 'tablet' breakpoint	PASSED	9.391
59	responsive/breakpoints.test.jsx	renders ProductCard safely at the 'desktop' breakpoint	PASSED	6.095

60	responsive/breakpoints.test.jsx	renders Home safely at the 'mobile' breakpoint	PASSED	296.441
61	responsive/breakpoints.test.jsx	renders Home safely at the 'tablet' breakpoint	PASSED	293.177
62	responsive/breakpoints.test.jsx	renders Home safely at the 'desktop' breakpoint	PASSED	292.299
63	utils/api.test.js	uses VITE_API_URL when configured	PASSED	2.867
64	utils/api.test.js	falls back to the implemented proxy/default API_BASE behaviour when no env override is set	PASSED	0.711
65	utils/api.test.js	returns the token from localStorage when present	PASSED	0.755
66	utils/api.test.js	prefers localStorage over sessionStorage for the stored token	PASSED	0.423
67	utils/api.test.js	falls back to sessionStorage when localStorage has no token	PASSED	0.389
68	utils/api.test.js	stores user data in the active auth storage	PASSED	0.633
69	utils/api.test.js	defaults to localStorage when storing a user without an active auth storage	PASSED	0.478
70	utils/api.test.js	attaches the Authorization header when a token exists	PASSED	2.019
71	utils/api.test.js	does not attach Authorization when no token exists	PASSED	0.542
72	utils/api.test.js	clears auth storage and dispatches auth:logout when authFetch receives 401	PASSED	0.860
73	utils/api.test.js	logout clears auth storage and dispatches	PASSED	0.515

		auth:logout		
74	utils/authRedirect.test.js	confirms and preserves the current path and search when navigating to login	PASSED	3.175
75	utils/authRedirect.test.js	does not navigate when the confirm dialog is cancelled	PASSED	0.469
76	utils/authRedirect.test.js	falls back to /home when location is missing	PASSED	0.456
77	forms/ChangePassword.test.jsx	marks the current password as required	PASSED	28.303
78	forms/ChangePassword.test.jsx	shows an error when the new password is too short	PASSED	697.703
79	forms/EditProduct.test.jsx	requires the title field before saving	PASSED	198.964
80	forms/EditProduct.test.jsx	shows saving state while a valid update is in progress	PASSED	105.458
81	forms/ForgotPassword.test.jsx	disables submission for an invalid email	PASSED	406.114
82	forms/ForgotPassword.test.jsx	shows success feedback after a valid request	PASSED	384.217
83	forms/Login.test.jsx	shows required field errors for missing email and password	PASSED	371.645
84	forms/Login.test.jsx	shows an error when the login API returns a failure	PASSED	570.779
85	forms/Login.test.jsx	keeps the submit button disabled until the form is valid	PASSED	543.087
86	forms/PublishProduct.test.jsx	requires a title before publishing	PASSED	618.707
87	forms/PublishProduct.test.jsx	marks a negative price as invalid	PASSED	541.693
88	forms/PublishProduct.test.jsx	requires a category when none is available or selected	PASSED	589.215

89	forms/PublishProduct.test.jsx	treats zero as an invalid price	SKIPPED	0.000
90	forms/PublishProduct.test.jsx	requires at least one uploaded image before publishing	SKIPPED	0.000
91	forms/Register.test.jsx	shows an error for an invalid university email	PASSED	1146.400
92	forms/Register.test.jsx	shows an error for a short password	PASSED	883.816
93	forms/Register.test.jsx	requires the email field before submission	PASSED	46.255
94	forms/Register.test.jsx	submits a valid form and shows success feedback	PASSED	1058.088
95	forms/ResetPassword.test.jsx	shows a password strength error for short passwords	PASSED	271.190
96	forms/ResetPassword.test.jsx	shows an error when the confirmation password does not match	PASSED	525.672

6. Reproducibility

From the project frontend directory, run: `npm run test:run`

For a machine-readable export of the same suite, run: `npx vitest run --reporter=json --outputFile vitest-report.json`